FIRE STARTING

Lighting, keeping and putting out a fire safely: outdoors, or in a fireplace or wood stove.

DOCUMENT TR-01
SERIES TRAINING
REVISION B // NOV 2026
STATUS PUBLIC ISSUE

GUIDE / VERIFY: General preparedness information, not a substitute for professional training, local fire rules, manufacturer instructions or medical care. Fire restrictions change with the weather and differ by place. **Check current local rules before you light any fire outdoors.**

1. BEFORE YOU LIGHT ANYTHING

CHECK THE RULES

- Look for burn bans and fire restrictions from your fire department, county, or the park or forest you are in. In dry weather they can change daily.
- Where a fire ring or pit is provided, use it.
- Never use gasoline, lighter fluid or other flammable liquids to start or revive a fire.

KEEP WITHIN REACH

- A bucket or large container of water.
- A shovel.
- A phone. If the fire gets out of the pit, call 911 right away.

PICK THE SPOT

- At least **15 feet** from tents, shrubs, trees and anything else that can burn.
- Clear overhead: branches at least **3 times the height** of the flames above the fire.
- Clear a circle at least **10 feet across** down to bare soil.
- No pit? Dig one about a foot deep and ring it with rocks or a metal fire ring.
- Stay out of windy, open areas.

(!) **NEVER LEAVE A FIRE UNATTENDED,** not even for a few minutes. Keep children and pets back from the fire ring.

2. HOW FIRE WORKS

A fire needs three things at once: **heat, fuel and air** (the fire triangle). Take away any one and it goes out. Most failed fires come down to one of three mistakes:

PROBLEM	WHAT YOU SEE	FIX
Fuel too big	Match burns out; sticks blacken but never catch.	Start smaller. Tinder first, then the thinnest kindling.
Fuel too wet	Hissing, heavy white smoke, flames die back.	Use wood that snaps cleanly. Split wet wood to reach the dry inside.
Too little air	Flames shrink and smoke builds when you add wood.	Leave gaps. Add wood a piece at a time; don't pile it on.

3. THREE KINDS OF FUEL

Gather everything **before** you light. A good rule is to collect about twice the kindling you think you need: once the tinder is burning, there is no time to go looking.

FUEL	SIZE	EXAMPLES	HOW MUCH
Tinder	Hair-thin to match-thin; catches from a spark or match.	Dry grass, dead pine needles, dry leaves, birch bark curls, wood shavings, cotton balls rubbed with petroleum jelly, store-bought fire starters.	Two bundles about the size of your fist.
Kindling	Sticks thinner than 1 inch: pencil to thumb thick.	Dead, dry twigs and split sticks.	Two good armloads.
Fuel wood	Wrist-thick and larger, split if possible.	Dead, dry wood, or firewood bought near where you burn it.	Enough for the whole fire.

- Good fuel is **dead and dry**: it snaps cleanly instead of bending. Green wood smokes and won't hold a flame.
- Don't cut live trees, and follow local rules about gathering wood.
- Never burn trash, plastics, aerosol cans or other pressurized containers, batteries, glass or aluminum.

4. BUILDING AND LIGHTING THE FIRE

FIRE LAY	HOW TO BUILD IT	BEST FOR
Teepee	Tinder in the center; lean kindling around it in a cone, with a gap on the side the wind comes from.	Starting a fire quickly; cooking.
Lean-to	Push a stick into the ground at an angle over the tinder, or use a log on the windward side; lean kindling against it.	Wind; cooking.
Log cabin	Build a small teepee, then stack pairs of sticks around it in a square, crossing at the corners.	A long, steady fire and a good bed of coals.

LIGHTING IT

1. Put the tinder bundle in place, loose enough for air to get in.
2. Kneel with your back to the wind and light the tinder low, on the side the wind comes from.
3. Feed the thinnest kindling one piece at a time as the flame grows.
4. If it falters, blow gently and steadily at the base.
5. Add thicker pieces only once the kindling is burning strongly.
6. Keep the fire only as big as you need.

WET OR WINDY CONDITIONS

- Look for dead branches still on standing trees, especially under thick evergreens. They are usually drier than wood on the ground.
- Split wet sticks; the center is often dry. Shave thin curls from the dry wood to use as tinder.
- Build on a platform of dry sticks or bark, not on wet ground.
- Carry fire starters (petroleum-jelly cotton balls in a sealed bag work well).
- Use a lean-to or shelter the fire behind a rock or log. In strong wind, don't build a fire at all.

5. WAYS TO LIGHT IT

Always carry **at least two** ways to start a fire, kept dry in a sealed bag or container.

METHOD	HOW TO USE IT	WATCH OUT FOR
Matches	Strike away from you and shield the flame with your body and hands. Hold the match head down a moment so the flame climbs the stick, then put it to the tinder.	Damp matches fail. Keep them in a waterproof case; stormproof matches stay lit in wind and rain.
Butane lighter	Light it and hold the flame at the base of the tinder.	Works poorly in cold. Carry it in an inside pocket to keep it warm. A wet lighter must be dried before it will spark.
Ferro rod (ferrocerium)	Rest the tip of the rod in the tinder. Hold the striker still at a steep angle against the rod and **pull the rod back** firmly. The sparks land in the tinder without knocking it apart.	Needs very fine, dry tinder. Still works when wet: wipe it dry first. Practice before you need it.
Magnifying lens	In bright sun, focus the light to the smallest dot you can on dark, dry tinder and hold it still until it smokes, then blow gently.	Useless on cloudy days or at night. Never leave a lens lying in the sun.

6. FIREWOOD AT HOME

DRY IT

- Dry, well-seasoned wood burns hotter and cleaner and makes less smoke and creosote. Aim for **less than 20% moisture**.
- Season softwood about **6 months** and hardwood **at least a year**.
- Split it, stack it off the ground, and **cover the top only**. Open sides let it dry.
- Check it with a wood moisture meter (an inexpensive tool). Dry wood is darker, weighs less and sounds hollow when two pieces are knocked together.

WHAT NOT TO BURN

- Painted, stained or pressure-treated wood, plywood, particle board, or anything with glue.
- Wet, rotted or moldy wood.
- Trash, cardboard, plastics, rubber or driftwood.

DON'T MOVE FIREWOOD

- Firewood can carry tree-killing insects and diseases. Buy it where you burn it: **within 10 miles is best; 50 miles is too far**.
- The exception is firewood certified as heat-treated, which is labeled as such.

7. INDOORS: FIREPLACES AND WOOD STOVES ONLY

- Burn wood indoors only in a fireplace or wood stove built for it, with a working chimney or vent.
- Have chimneys and vents **inspected at least once a year** and cleaned as needed.
- Keep anything that can burn at least **3 feet** from the fireplace or stove. Use a fireplace screen.
- Let ashes cool, then put them in a **metal container with a tight lid**, outside, at least **10 feet** from the house. Never empty ashes into a trash can.
- Have working smoke alarms and **carbon monoxide alarms** near sleeping areas. Test them monthly.

> ! **CARBON MONOXIDE KILLS WITHOUT WARNING.** It has no color or smell. Never use a grill, camp stove, generator or any charcoal- or gasoline-burning device inside a home, basement, garage or camper, even with the doors or windows open, or outside within **20 feet** of a window, door or vent. If a CO alarm sounds, or anyone feels dizzy, sick or confused, get everyone into fresh air and call 911.

8. PUTTING IT OUT: DROWN, STIR, DROWN, FEEL

1. **DROWN** the fire with plenty of water until the hissing stops. Dirt or sand alone is not enough.
2. **STIR** the water into the ashes and embers with a shovel. Scrape burnt sticks and logs.
3. **DROWN** it again until everything is wet and cool.
4. **FEEL**: hold the back of your hand just above the ashes. If you feel any heat, start again.

> **If it's too hot to touch, it's too hot to leave.** Buried coals can stay hot for many hours and restart a fire after you've gone. Never bury a fire to put it out.

9. MINOR BURNS

1. Cool the burn under cool running water for **at least 10 minutes**. Many burn specialists advise up to 20.
2. Remove rings, watches and tight clothing near the burn before it swells.
3. Cover it loosely with a clean, non-stick dressing.
4. Don't use ice, butter or ointments, and don't break blisters.

> ! **CALL 911** for large or deep burns; burns to the face, hands, feet, groin or major joints; electrical or chemical burns; or if anyone has trouble breathing or may have breathed in smoke.

10. TRAINING RECORD

Practice before you need it, in good weather and in bad. Record each attempt.

DATE	CONDITIONS	METHOD	TIME TO FLAME	NOTES

SOURCES / FURTHER REFERENCE

1. Smokey Bear (USDA Forest Service, National Association of State Foresters, Ad Council): How to Build Your Campfire; Campfire Safety. smokeybear.com
2. U.S. Environmental Protection Agency, Burn Wise: Best Wood-Burning Practices. epa.gov/burnwise
3. National Fire Protection Association: Heating Safety Tips; Educational Messages Desk Reference. nfpa.org
4. Centers for Disease Control and Prevention: Avoiding Carbon Monoxide Poisoning. cdc.gov/carbon-monoxide
5. Don't Move Firewood (The Nature Conservancy); USDA Animal and Plant Health Inspection Service. dontmovefirewood.org
6. American Red Cross: Burn Cooling Advisory (2015 AHA / Red Cross First Aid Guidelines). redcross.org
7. FEMA: Ready.gov. ready.gov

IMPORTANT: Fire restrictions, equipment instructions and first-aid recommendations change. This document reflects the sources above as of November 2026. Your local fire department and land manager have the final word on where and when you may burn.

LAST PRACTICED: _____ SUPPLIES CHECKED: _____